Enterprise Office Network Design and Implementation Using Cisco Packet Tracer (CCNA Practice)

Description

This project demonstrates the design and implementation of a small enterprise office network using Cisco Packet Tracer. The main objective is to build a real-world like network that can be used in an office environment where multiple departments communicate securely and efficiently.

The network includes different departments such as HR, Finance, IT, and Administration. Each department is separated using VLANs to improve security and reduce unnecessary network traffic.

The project focuses on core CCNA concepts such as:

- VLAN configuration
- Inter-VLAN routing
- DHCP configuration
- Routing protocols (OSPF)
- Access Control Lists (ACL)
- IP addressing and subnetting
- Network troubleshooting

Objectives

The main goals of this project were:

- To design a structured and scalable office network
- To divide the network into multiple departments using VLANs
- To enable communication between different VLANs using routing
- To configure DHCP for automatic IP assignment
- To improve security using ACL rules
- To ensure stable and efficient communication between all devices
- To gain hands-on experience with Cisco networking devices

Tools and Technologies Used

- Cisco Packet Tracer (Network Simulation Tool)
- Cisco Routers (Layer 3 routing)
- Cisco Switches (Layer 2 switching)
- End devices (PCs, Servers)
- Ethernet cables (connections)

- IPv4 addressing scheme
- VLAN technology
- OSPF routing protocol
- DHCP server configuration
- ACL (Access Control List)

Network Design Overview

The network was designed in a hierarchical structure:

- Core Layer: Router (handles routing between networks)
- Distribution Layer: Switches (connect departments)
- Access Layer: PCs and end devices

Each department was separated logically using VLANs:

- VLAN 10 → HR Department
- VLAN 20 → Finance Department
- VLAN 30 → IT Department
- VLAN 40 → Admin Department

This separation ensures:

- Better security
- Less network congestion
- Easier management

IP Addressing Scheme

A private IPv4 addressing scheme was used:

- VLAN 10 (HR): 192.168.10.0/24
- VLAN 20 (Finance): 192.168.20.0/24
- VLAN 30 (IT): 192.168.30.0/24
- VLAN 40 (Admin): 192.168.40.0/24

Each VLAN was assigned a gateway IP through router sub-interfaces.

DHCP was configured so that devices automatically receive:

- IP Address
- Subnet Mask

- Default Gateway
- DNS (optional)
-

Step-by-Step Implementation (What I did in the project)

Step 1: Network Topology Creation

I started by opening Cisco Packet Tracer and designing the network layout. I placed routers, switches, and multiple PCs representing different departments.

Step 2: Device Connection

All devices were connected using Ethernet cables:

- PCs connected to switches
- Switches connected to router
- Proper port selection was ensured

Step 3: VLAN Creation

I created separate VLANs on the switches:

- VLAN 10 for HR
- VLAN 20 for Finance
- VLAN 30 for IT
- VLAN 40 for Admin

Then I assigned switch ports to correct VLANs so each PC belongs to its department.

Step 4: Trunk Configuration

I configured trunk links between switch and router so that multiple VLAN traffic can pass through a single connection.

Step 5: Inter-VLAN Routing

I configured router sub-interfaces for each VLAN:

- Each sub-interface was assigned an IP address
- Encapsulation (dot1Q) was configured
- This allowed communication between different VLANs

Step 6: DHCP Configuration

I configured DHCP pools on the router for each VLAN.

This allowed:

- Automatic IP assignment
- No manual configuration on PCs
- Reduced configuration errors

Step 7: Routing Configuration (OSPF)

I configured OSPF routing protocol on the router.

This helped in:

- Sharing routing information automatically
- Finding best paths for data transfer
- Making the network scalable

Step 8: Security Implementation (ACL)

I applied Access Control Lists (ACLs) to control traffic.

Examples:

- HR should not access Finance data
- IT department has full access
- Restricted unnecessary communication

This improved network security.

Step 9: Testing the Network

I tested the network using:

- Ping command
- Traceroute
- Simulation mode in Packet Tracer

All departments were checked for:

- IP assignment
- Connectivity
- Routing accuracy

Step 10: Troubleshooting

During testing, I fixed issues such as:

- Wrong VLAN assignment
- IP configuration errors
- Routing misconfigurations
- Cable connection mistakes

Main Features of the Project

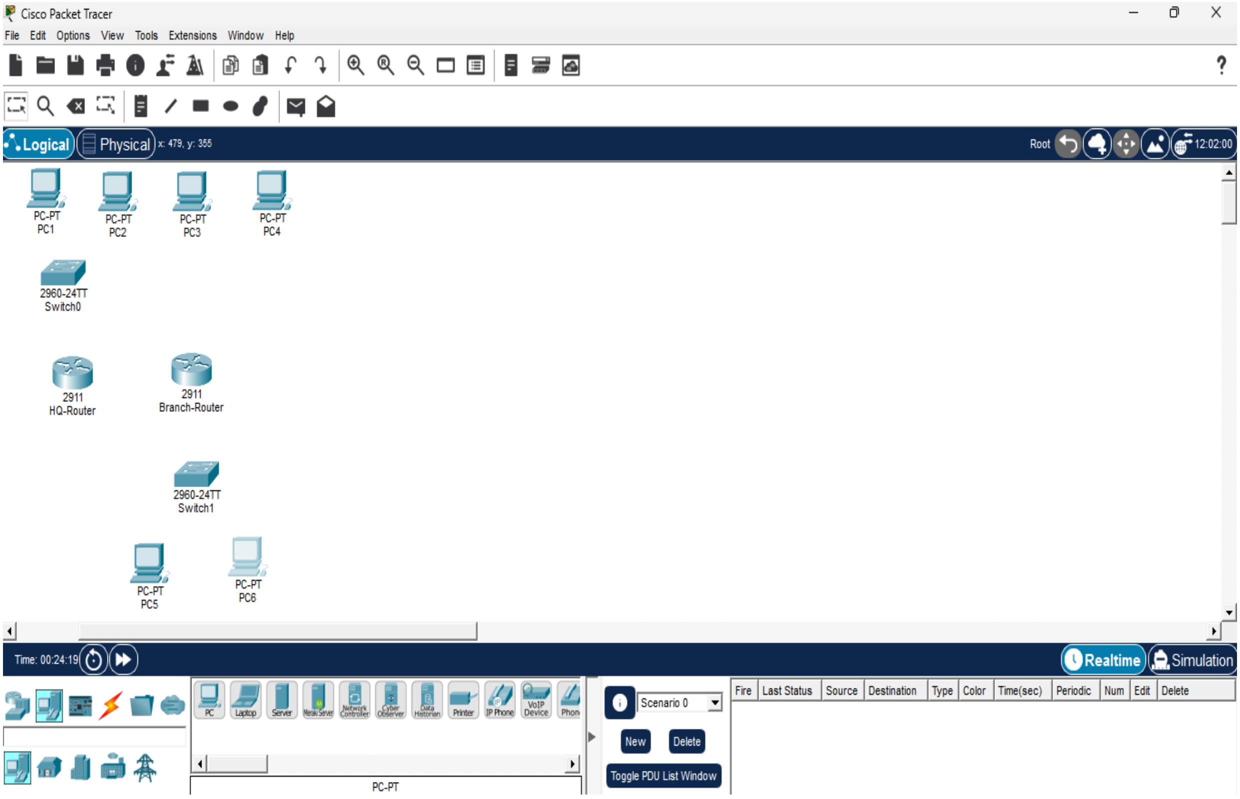
- VLAN-based network segmentation
- Secure inter-department communication
- Automatic IP assignment using DHCP
- Dynamic routing using OSPF
- Network security using ACL
- Real-world enterprise network simulation

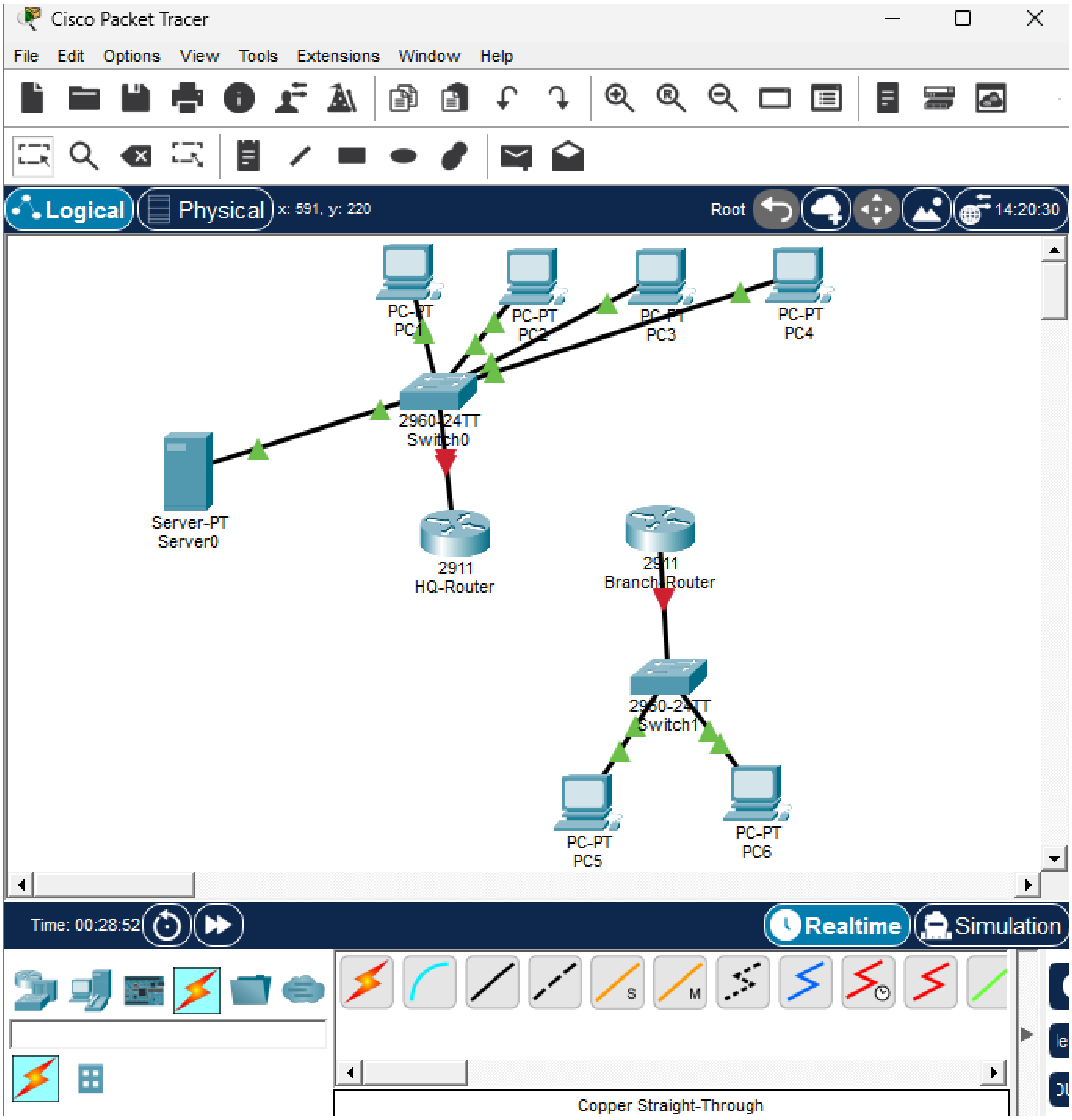
The project was successfully completed. The network worked as expected with:

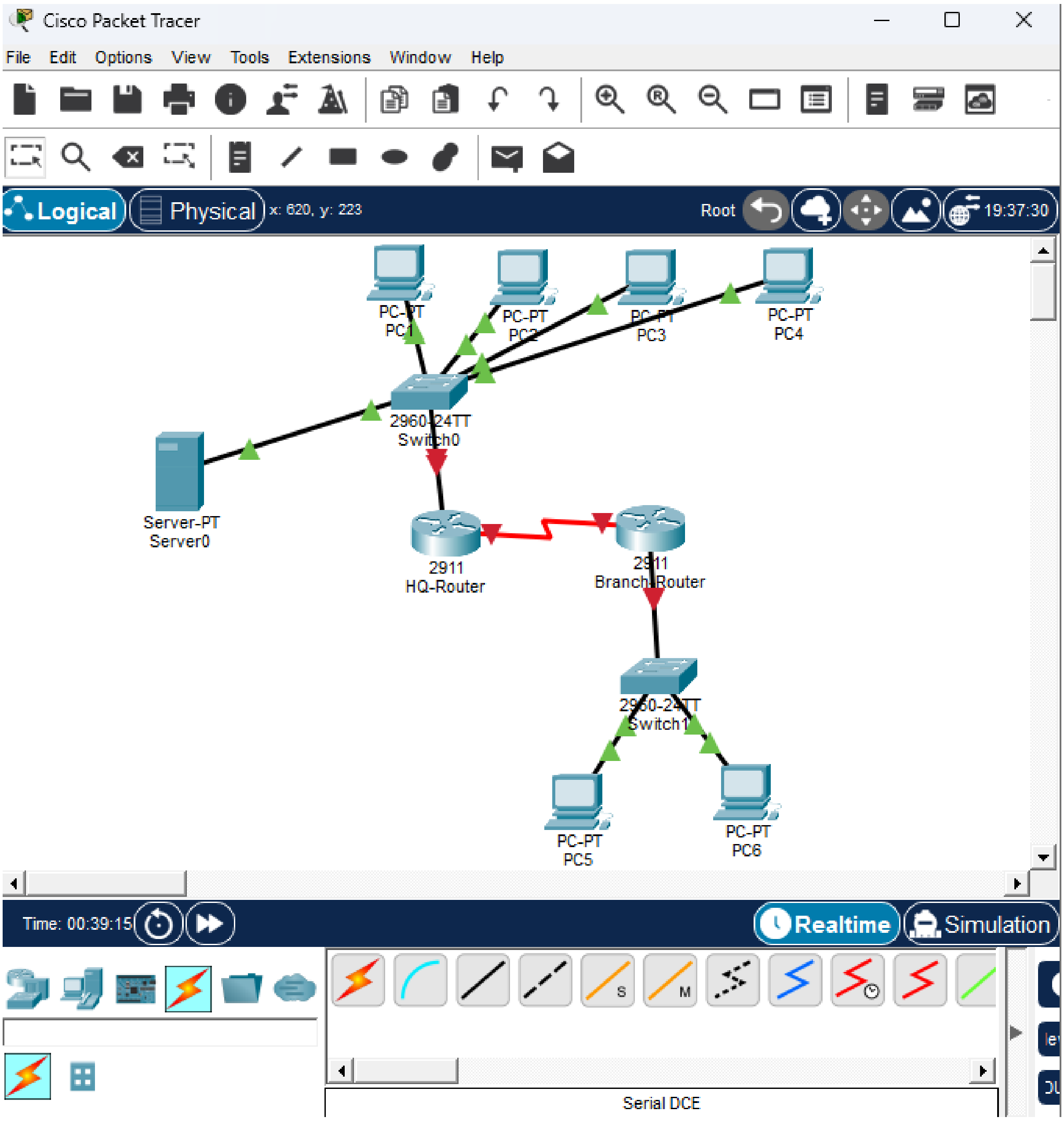
- Proper communication between allowed departments
- Secure separation between VLANs
- Automatic IP allocation
- Stable routing and connectivity

The system behaves like a real office network environment.

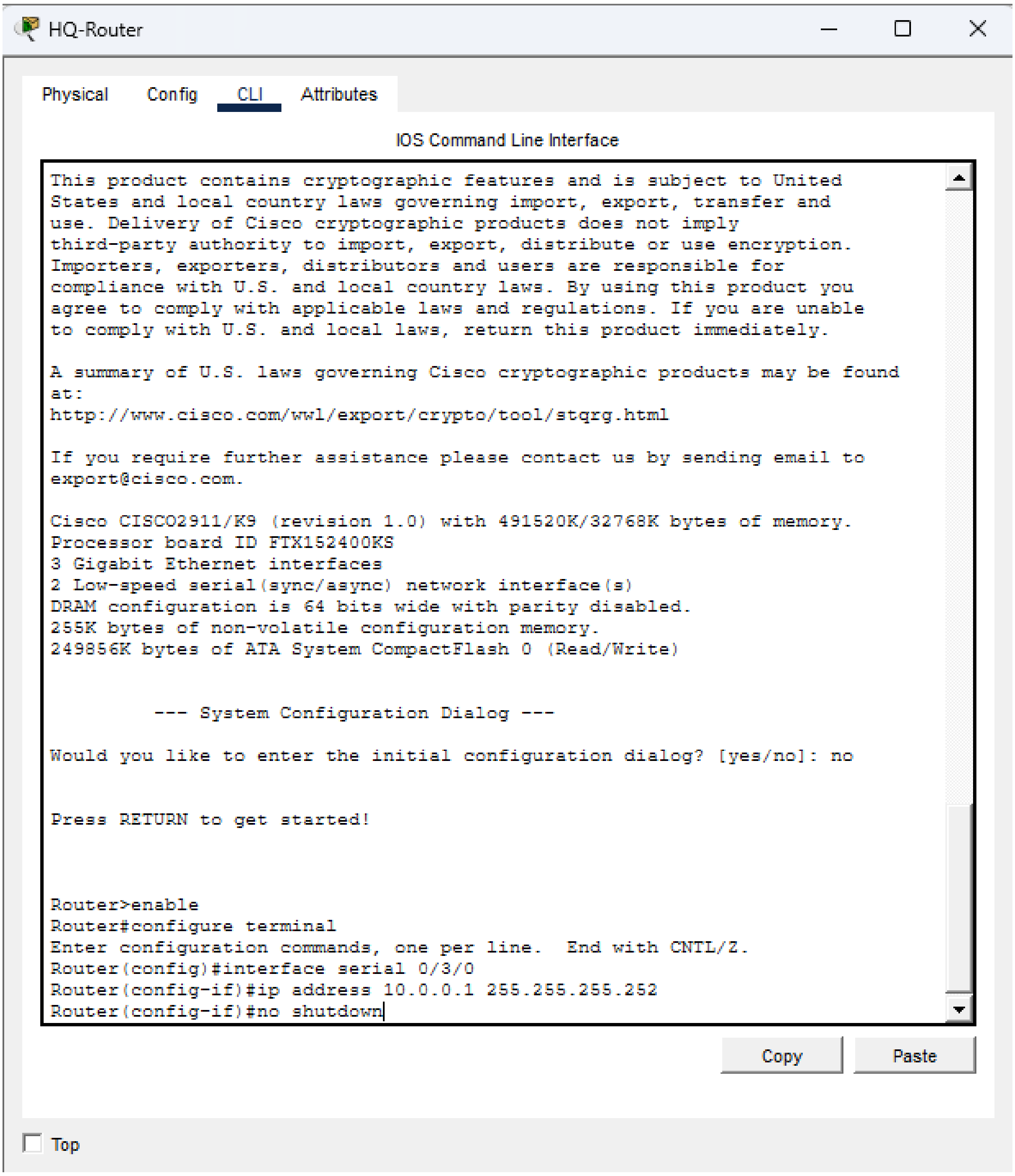
Screenshots:

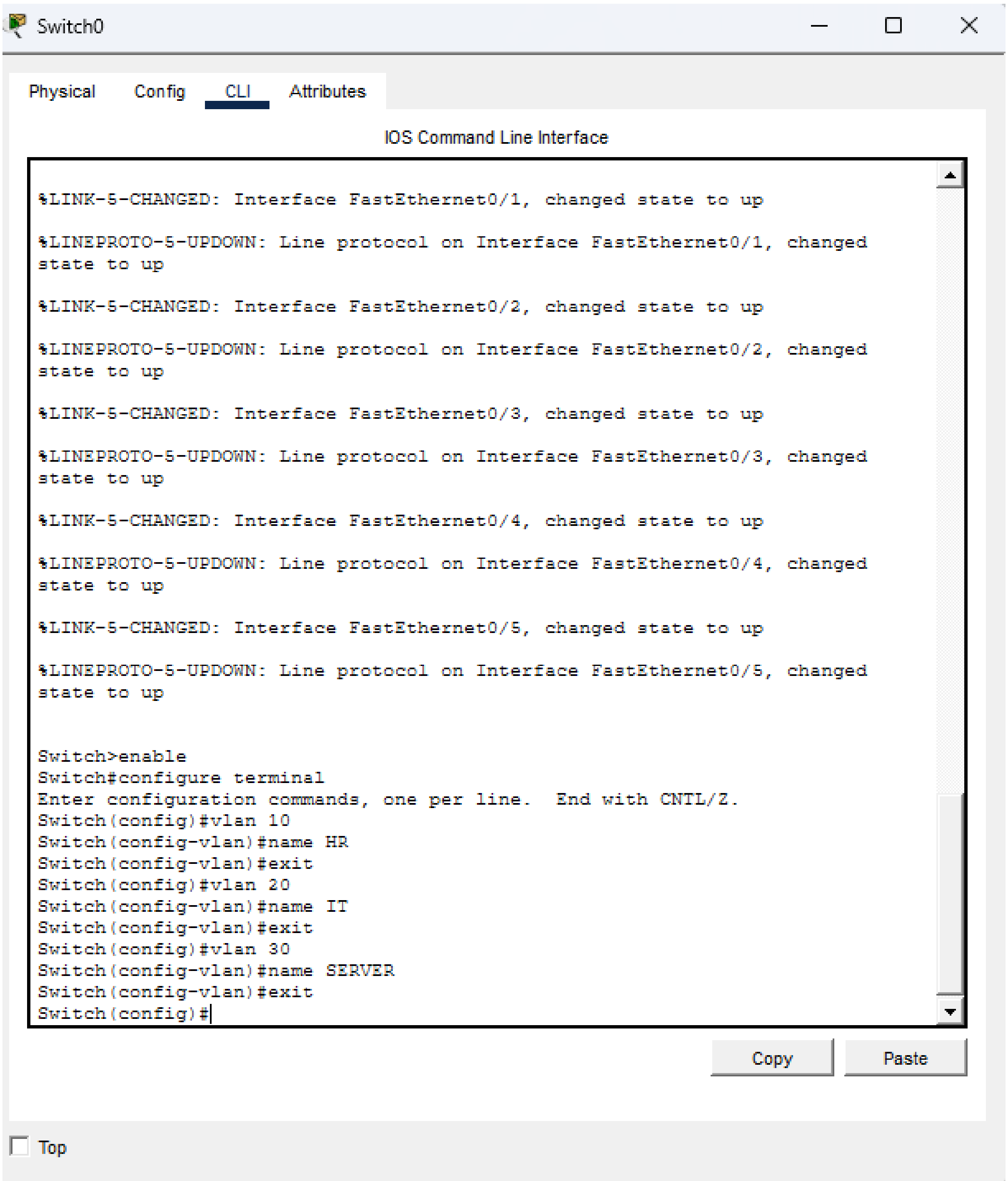

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 479, y: 355
Root
12:02:00
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
2911
HQ-Router
2911
Branch-Router
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6
Time: 00:24:19
Realtime
Simulation
PC
Laptop
Server
Printer
IP Phone
VoIP Device
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
PC-PT

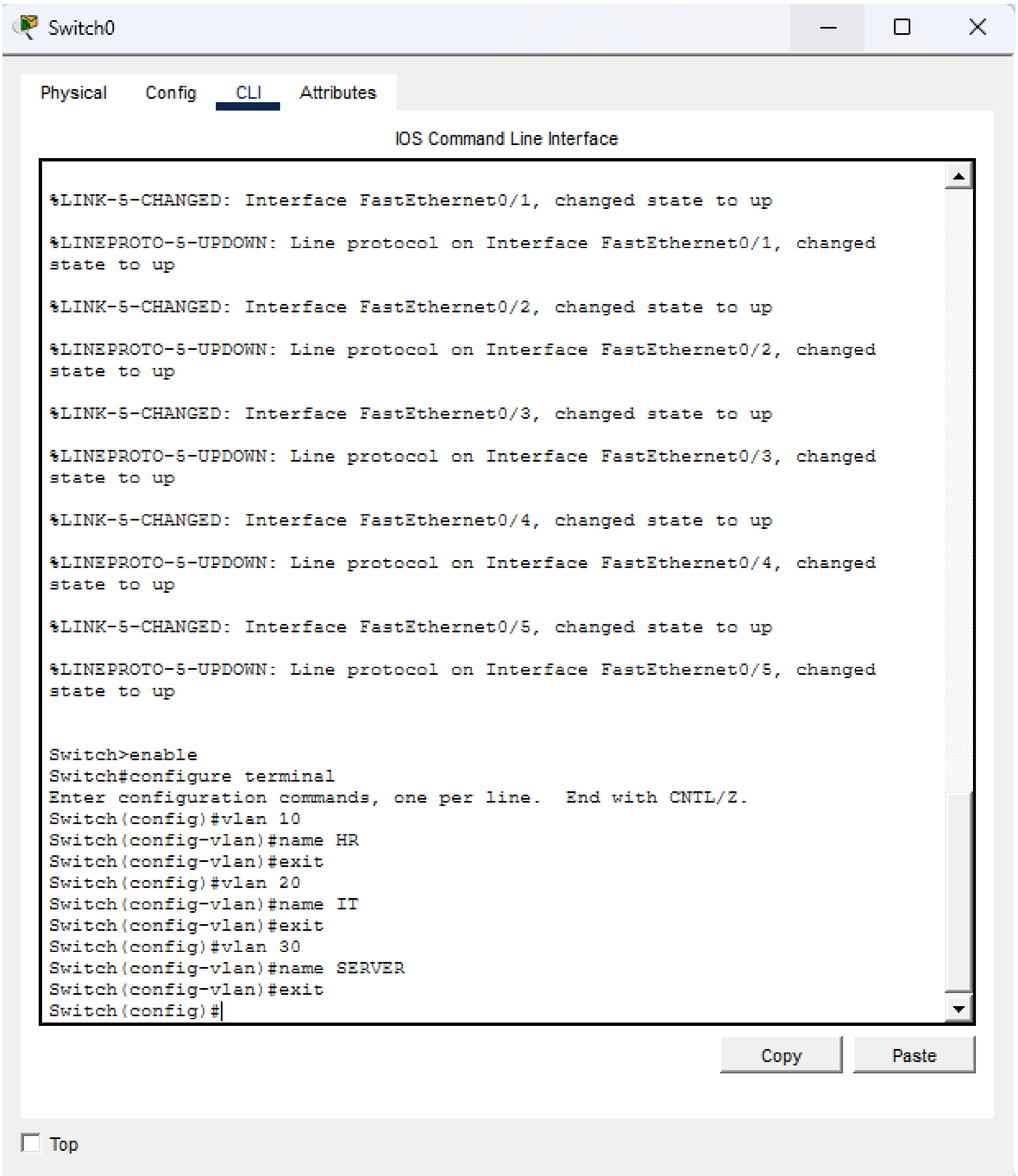

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 591, y: 220
Root
14:20:30
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
Server-PT
Server0
2911
HQ-Router
2911
Branch-Router
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6
Time: 00:28:52
Realtime
Simulation
Copper Straight-Through



Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 620, y: 223
Root
19:37:30
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
Server-PT
Server0
2911
HQ-Router
2911
Branch Router
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6
Time: 00:39:15
Realtime
Simulation
Serial DCE


HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found
at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Copy
Paste
Top


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name SERVER
Switch(config-vlan)#exit
Switch(config)#
Copy
Paste
Top



Switch0

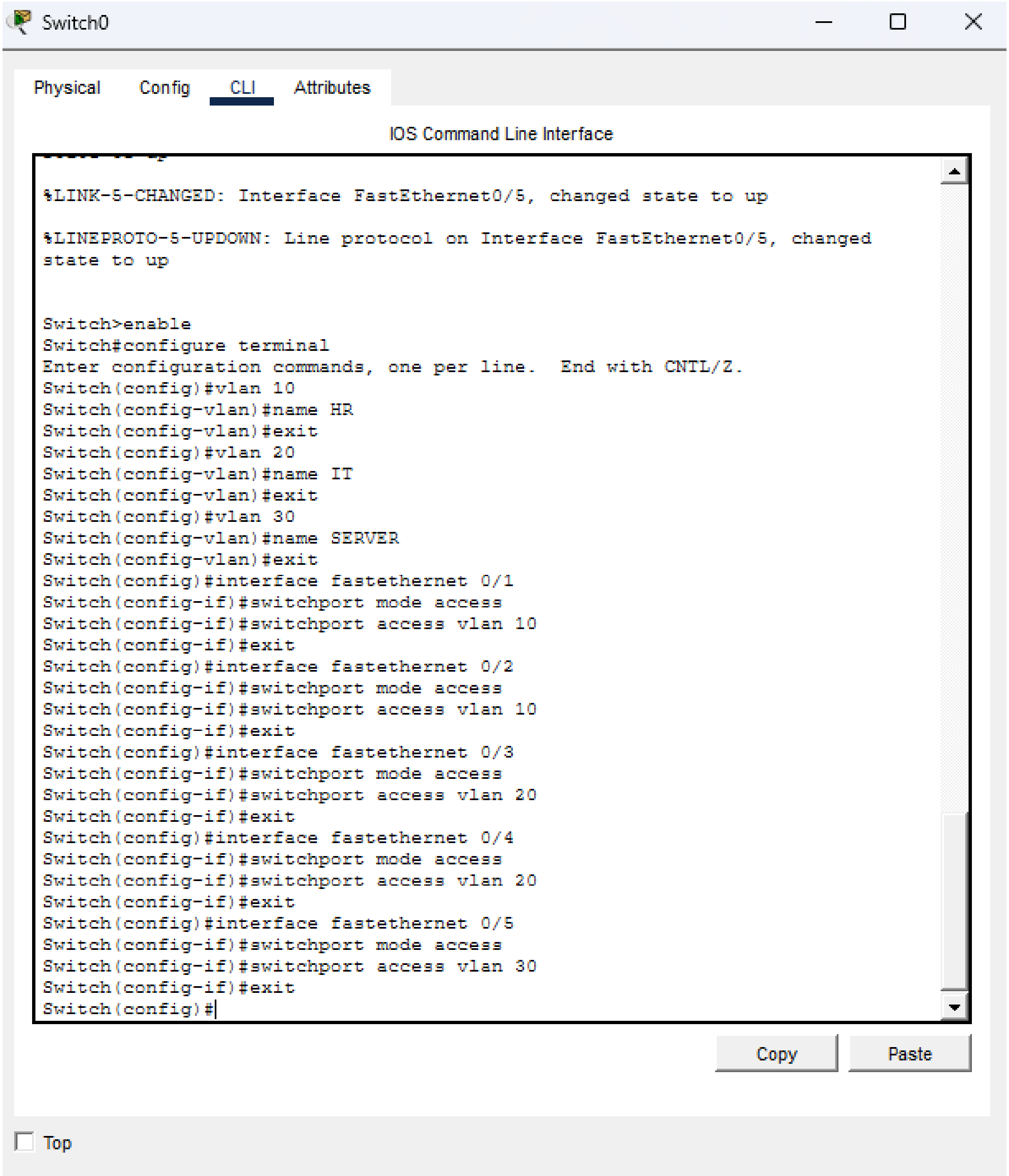
Physical Config CLI Attributes

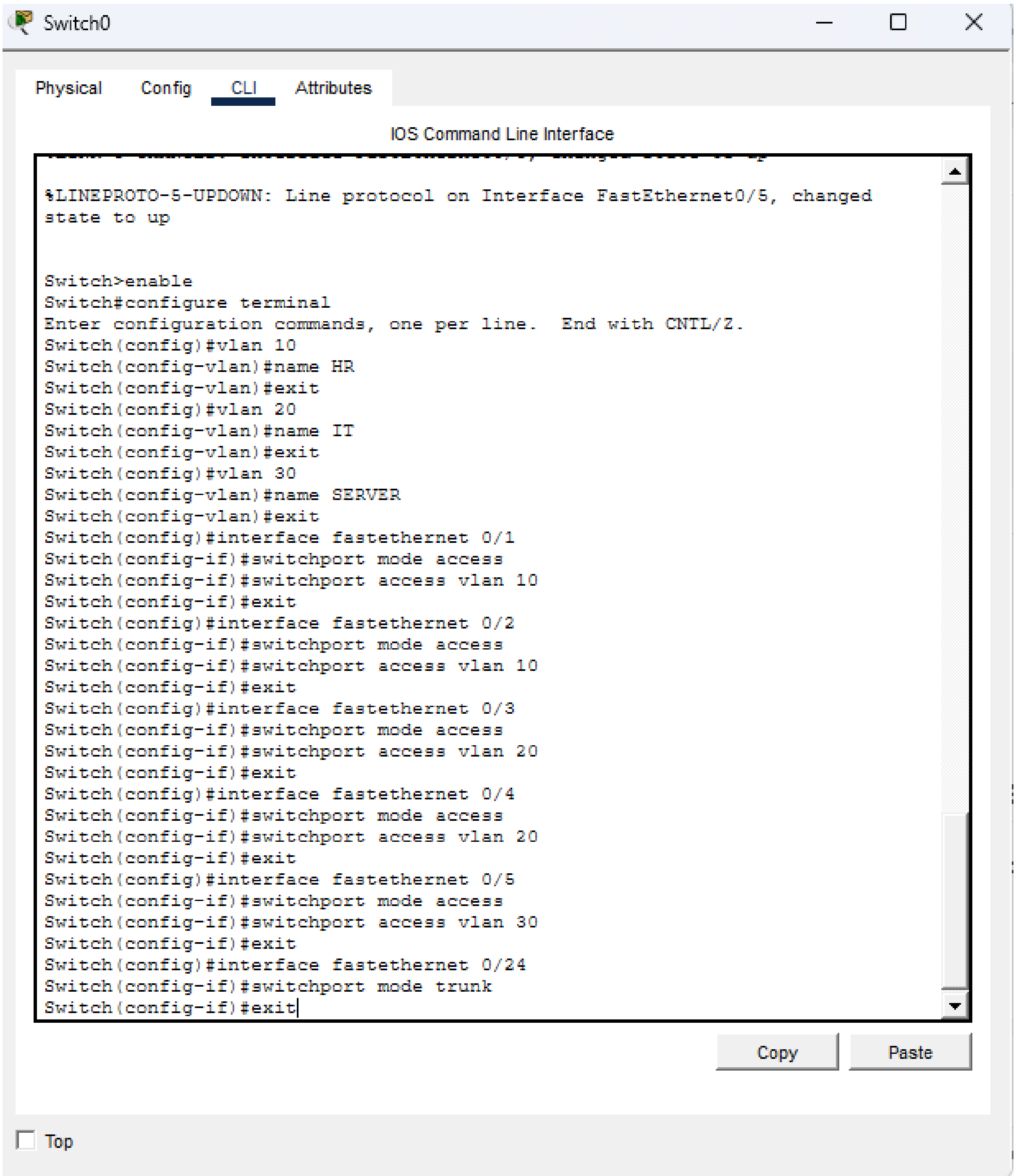
IOS Command Line Interface

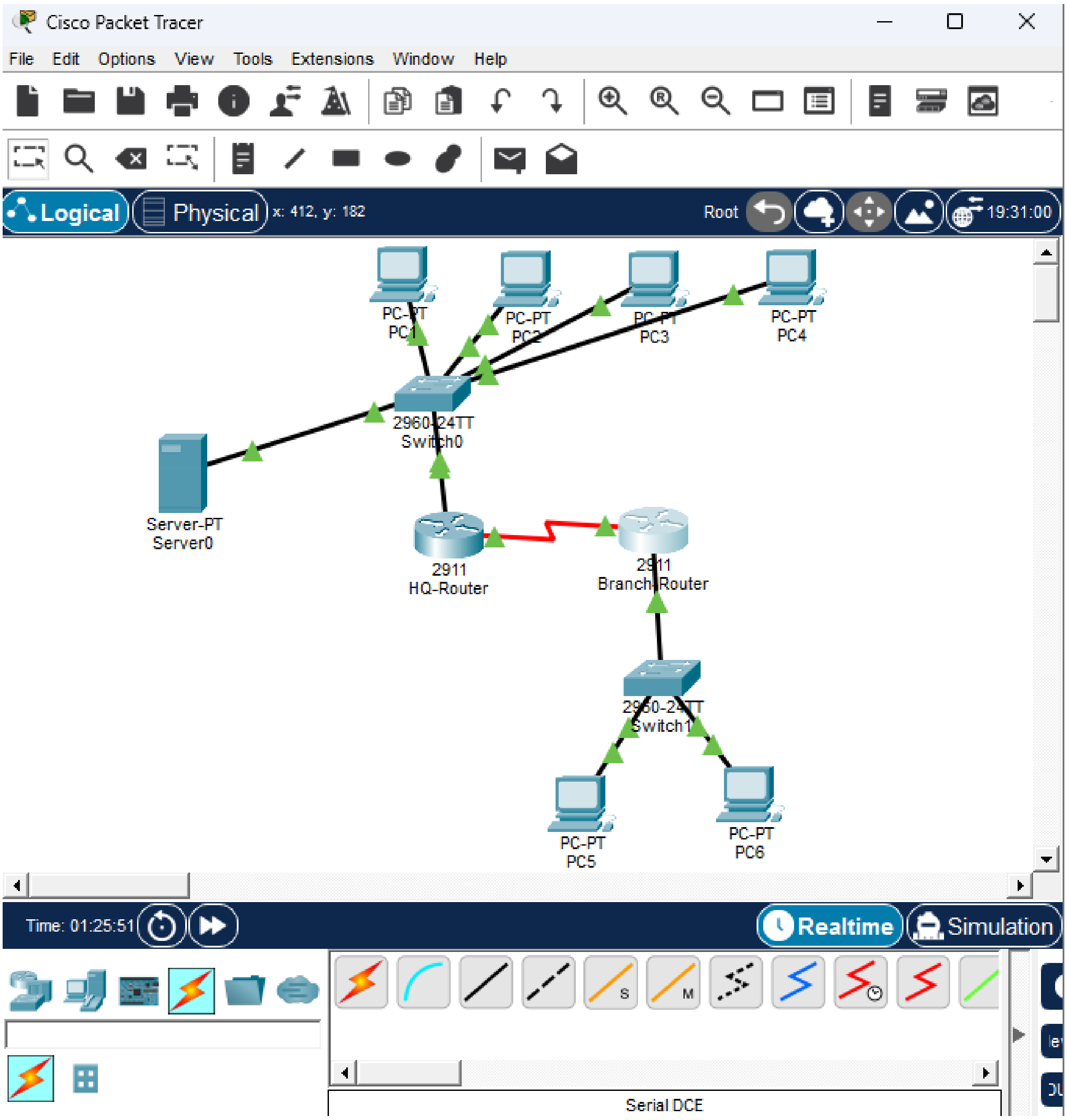
```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed
state to up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name SERVER
Switch(config-vlan)#exit
Switch(config)#
```

Copy Paste

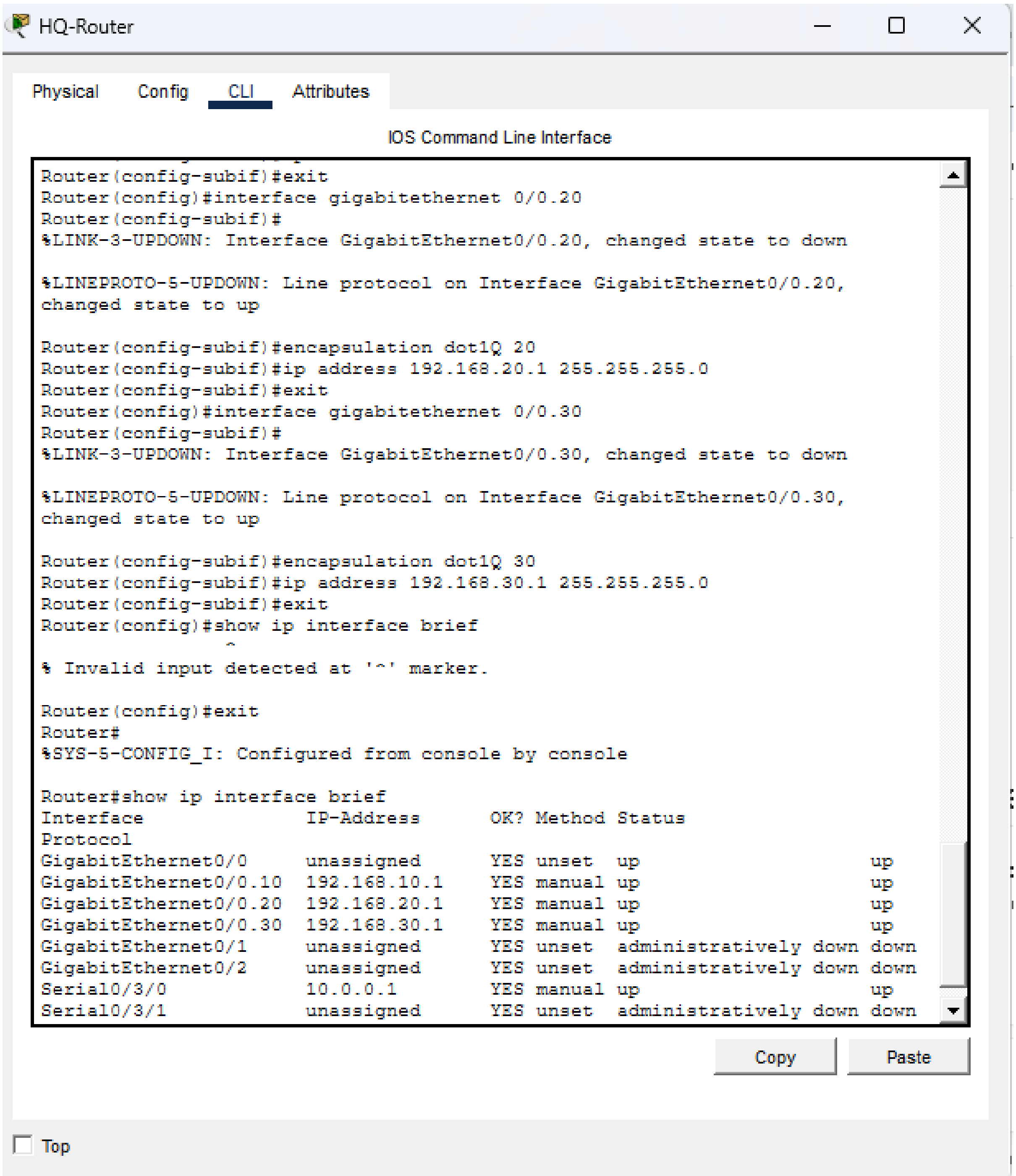
Top


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name SERVER
Switch(config-vlan)#exit
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
Copy
Paste
Top

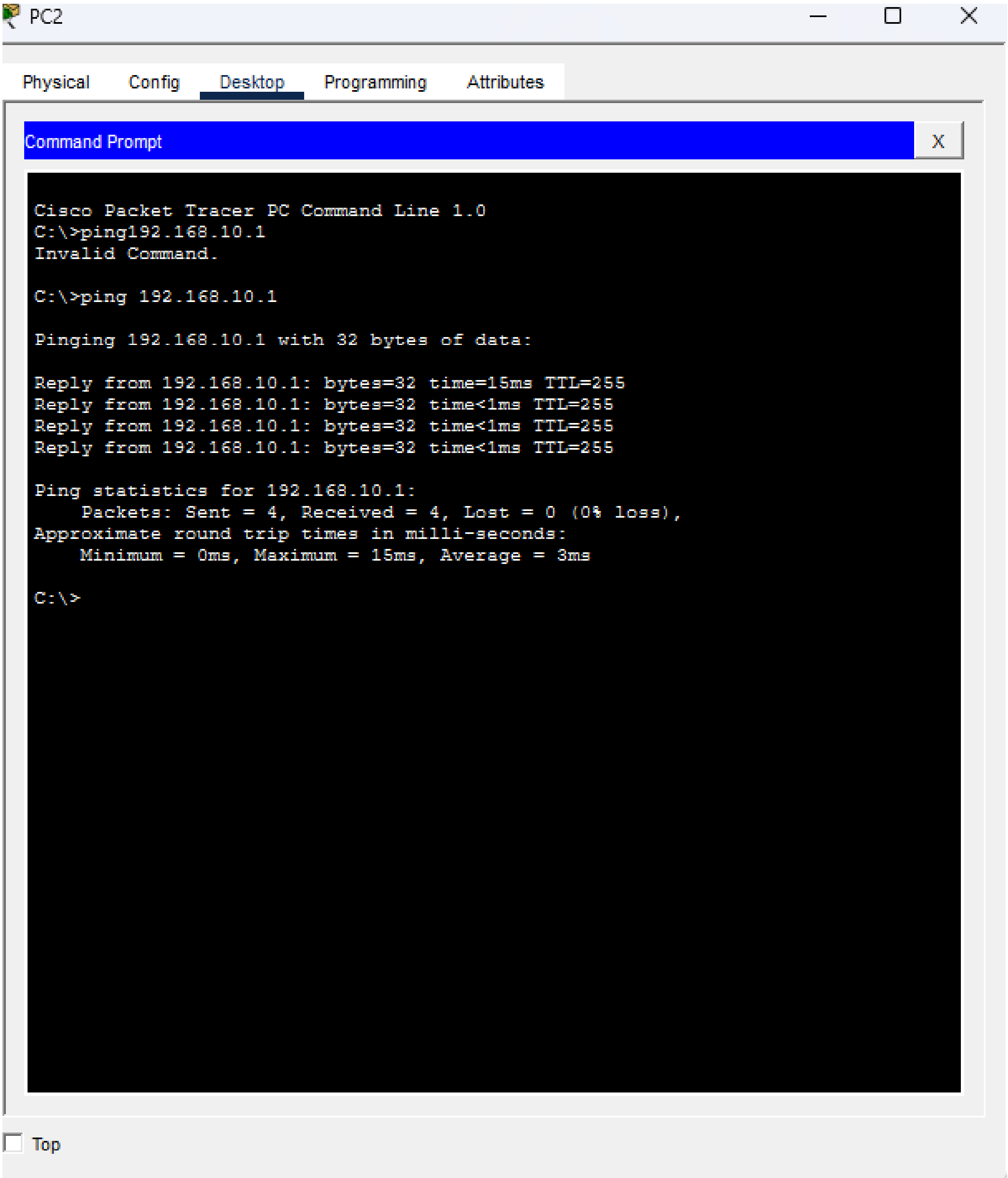

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name SERVER
Switch(config-vlan)#exit
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Copy
Paste
Top

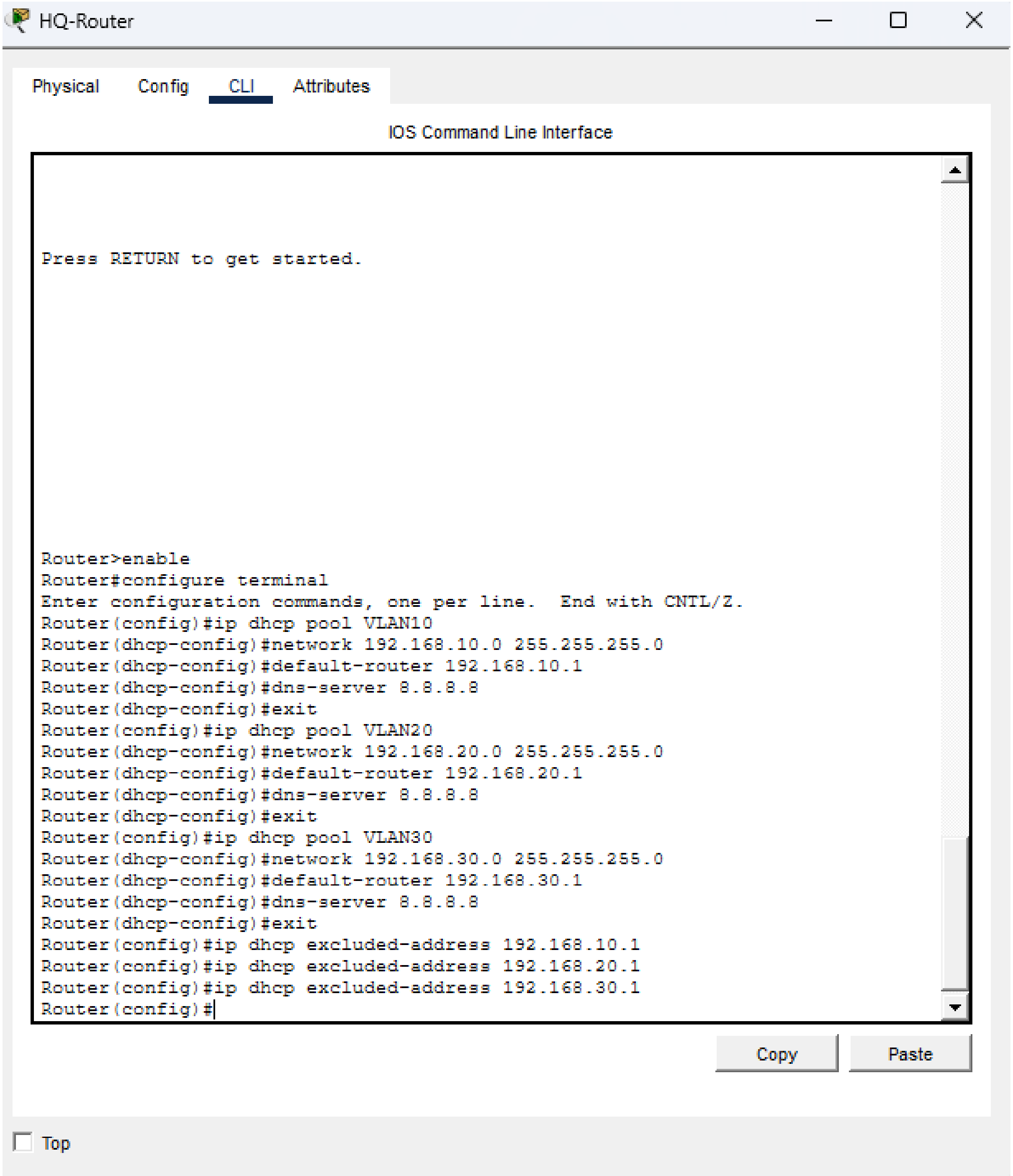


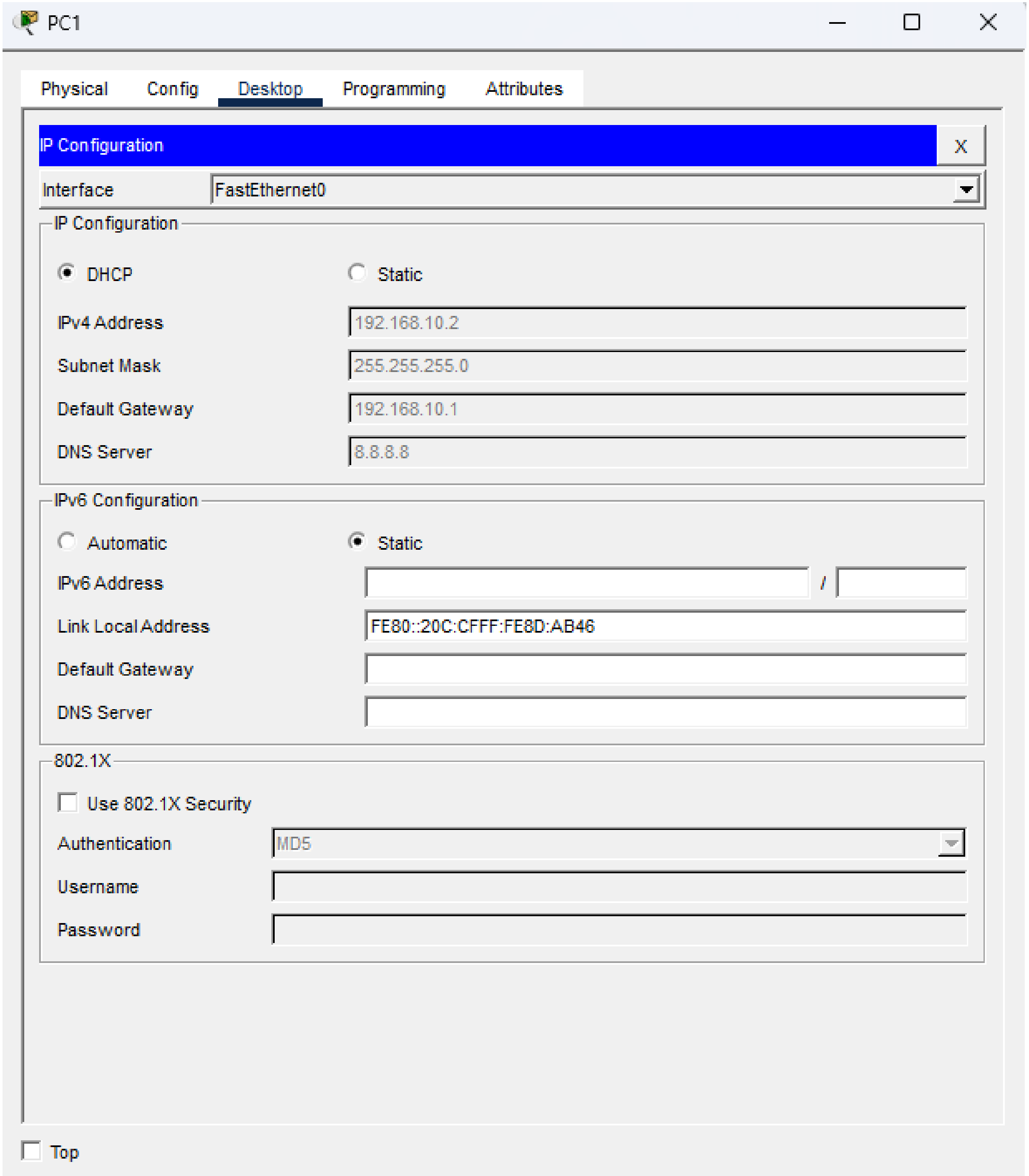
Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 412, y: 182
Root
19:31:00
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
Server-PT
Server0
2911
HQ-Router
2911
Branch-Router
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6
Time: 01:25:51
Realtime
Simulation
Serial DCE

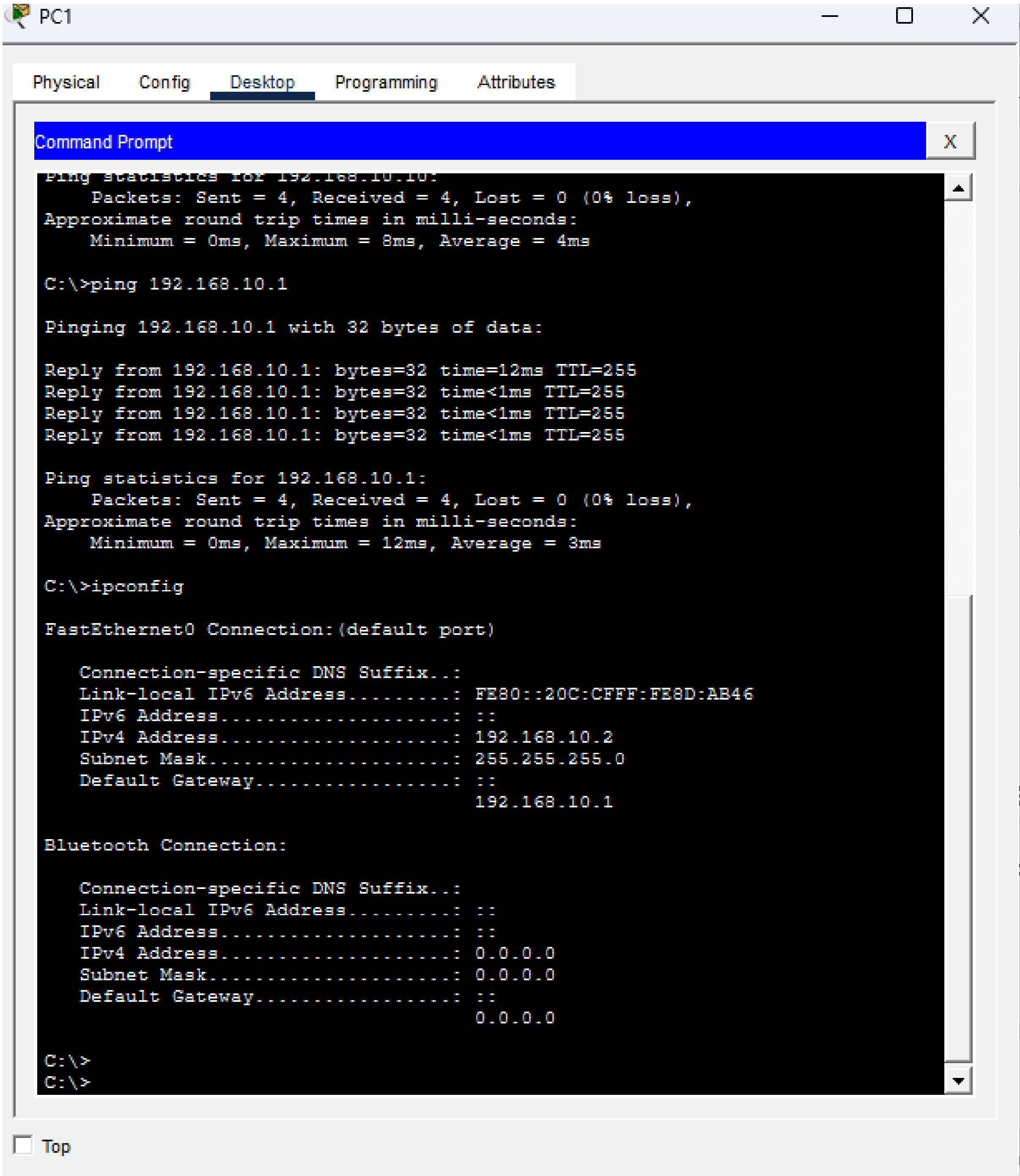


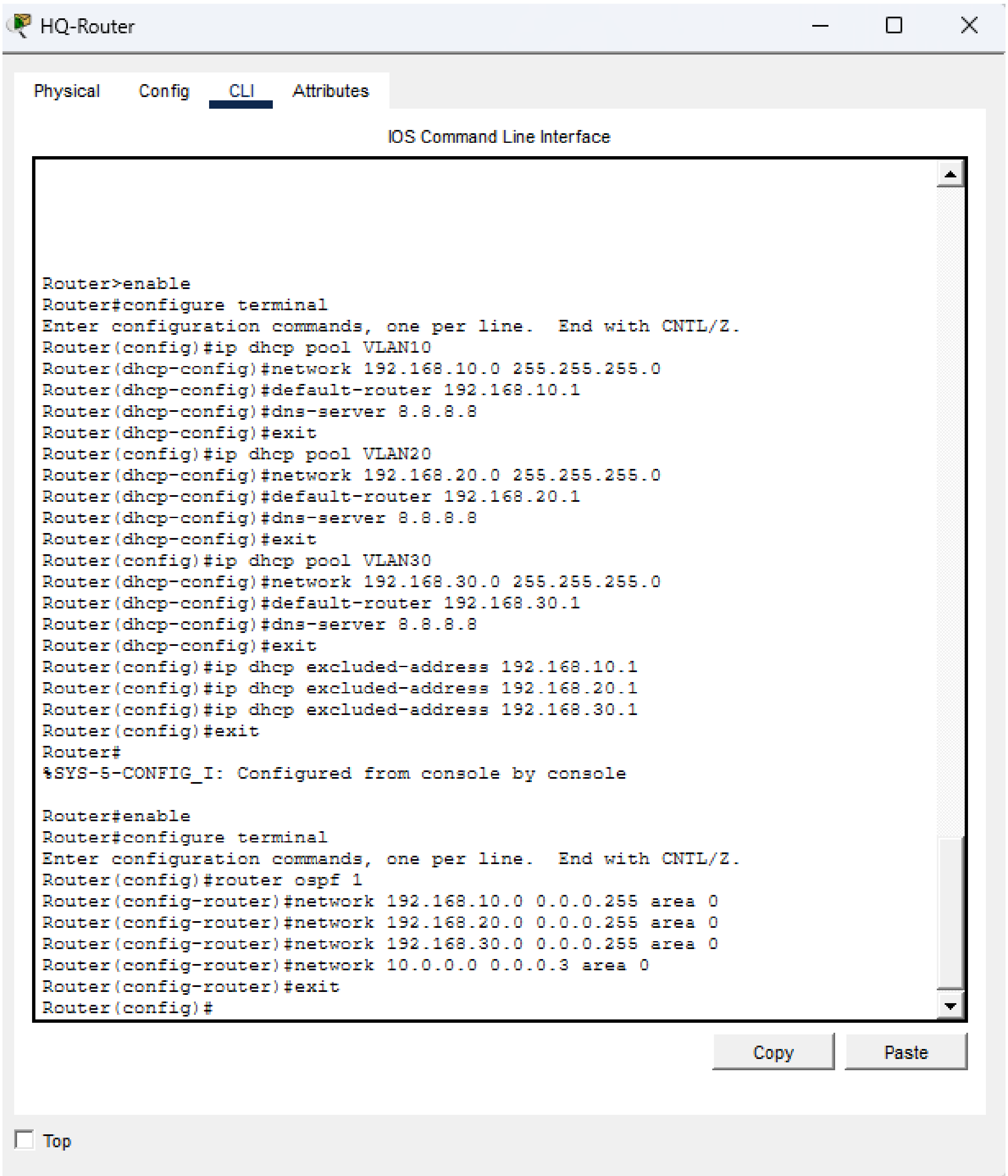

```
Router(config-subif)#exit
Router(config)#interface gigabitethernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20,
changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitethernet 0/0.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30,
changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#show ip interface brief
                ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.1        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
```

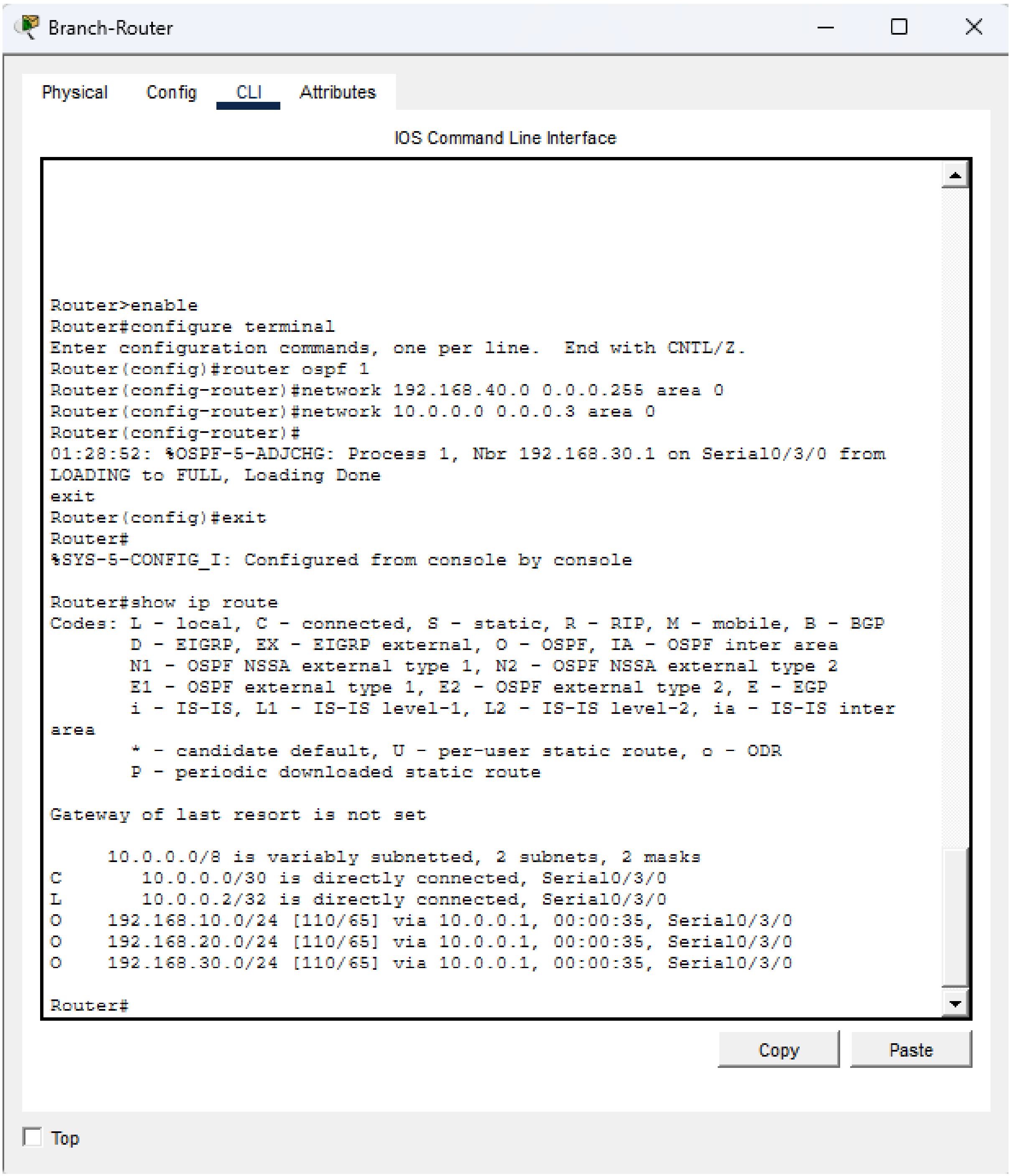

PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping192.168.10.1
Invalid Command.
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=15ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 3ms
C:\>
Top


HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool VLAN10
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN20
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN30
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1
Router(config)#
Copy
Paste
Top

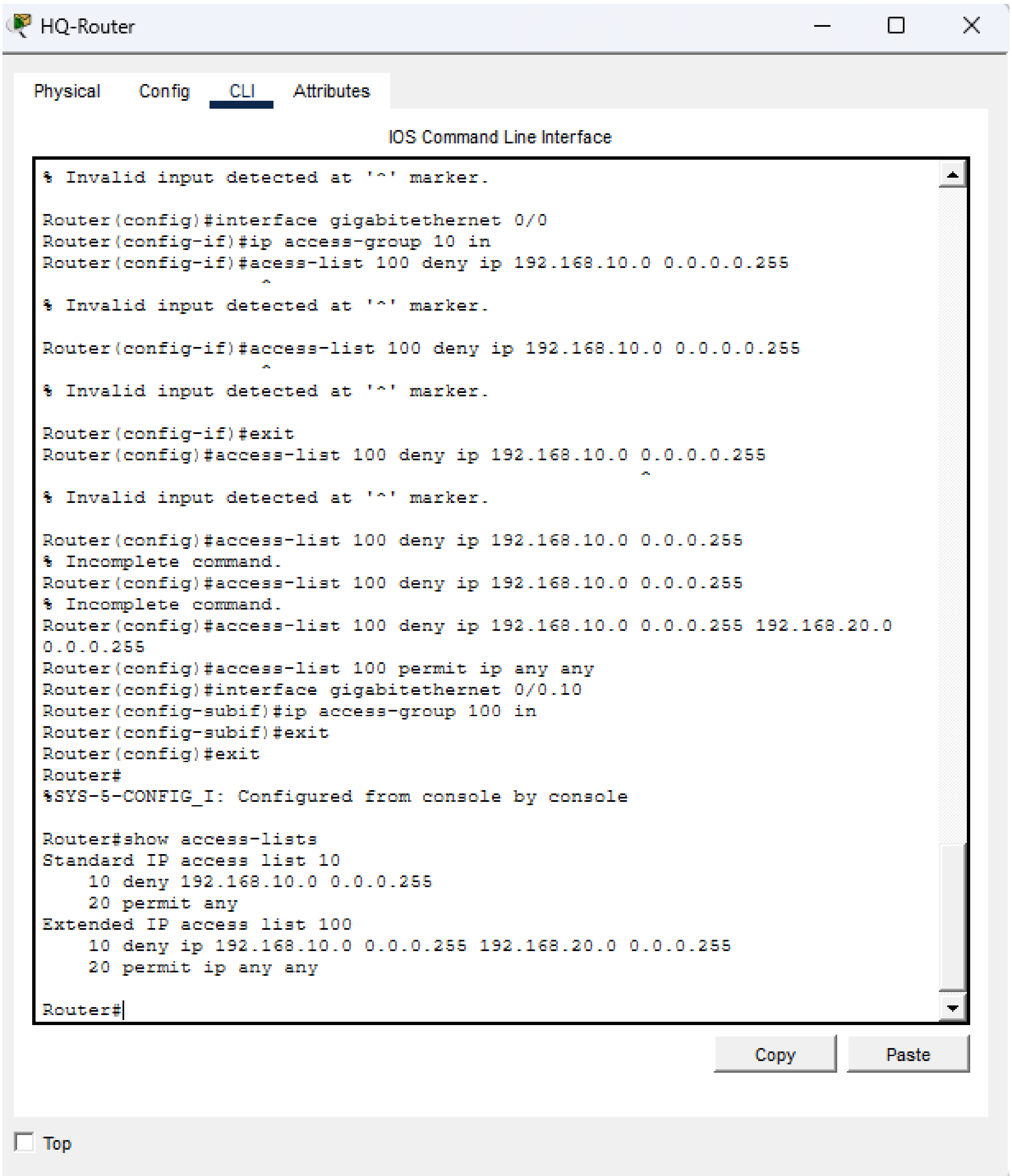

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20C:CFFF:FE8D:AB46
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

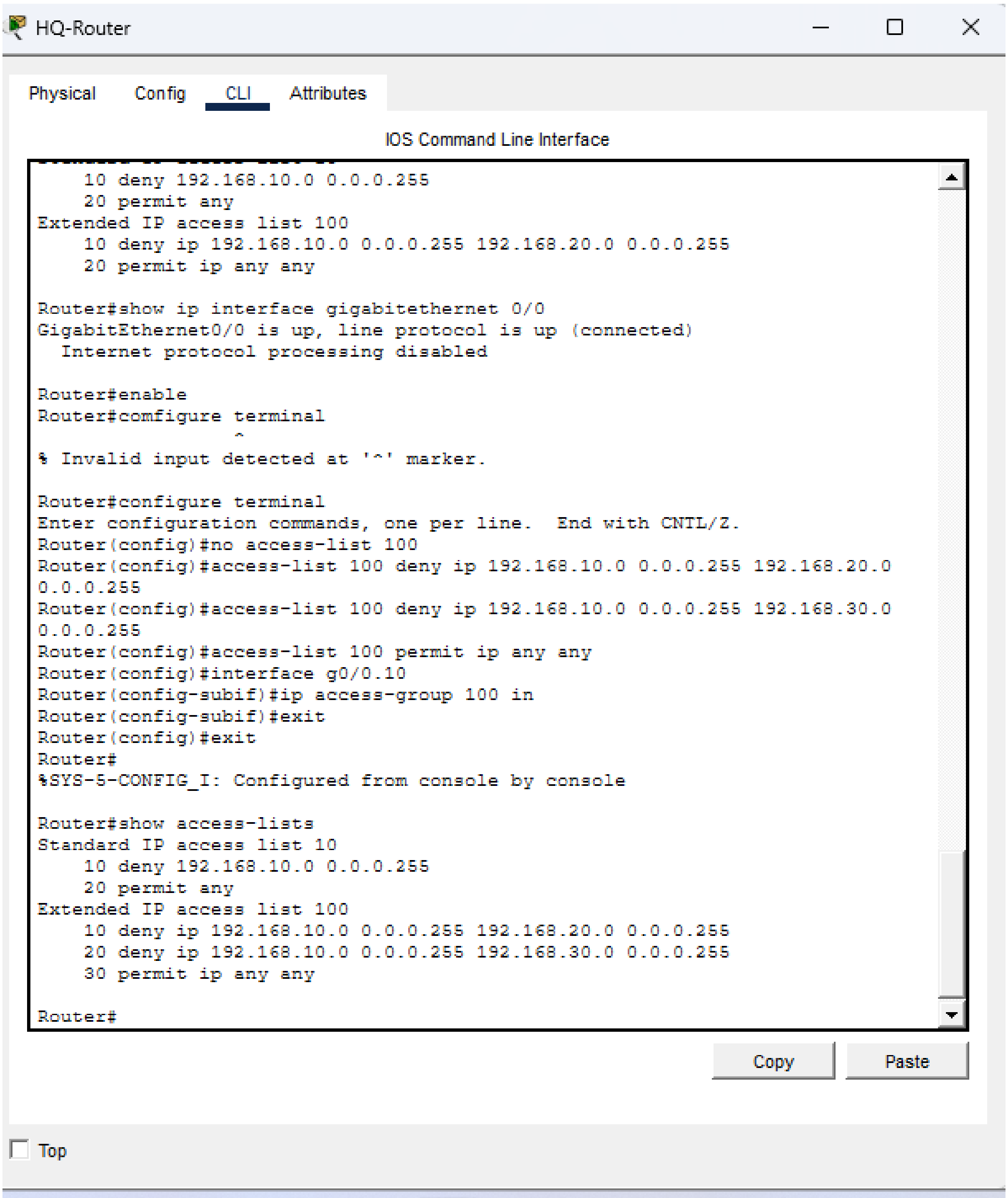

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 8ms, Average = 4ms
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=12ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 3ms
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: FE80::20C:CFFF:FE8D:AB46
IPv6 Address....................: ::
IPv4 Address....................: 192.168.10.2
Subnet Mask.....................: 255.255.255.0
Default Gateway.................: ::
192.168.10.1
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address....................: ::
IPv4 Address....................: 0.0.0.0
Subnet Mask.....................: 0.0.0.0
Default Gateway.................: ::
0.0.0.0
C:\>
C:\>
Top

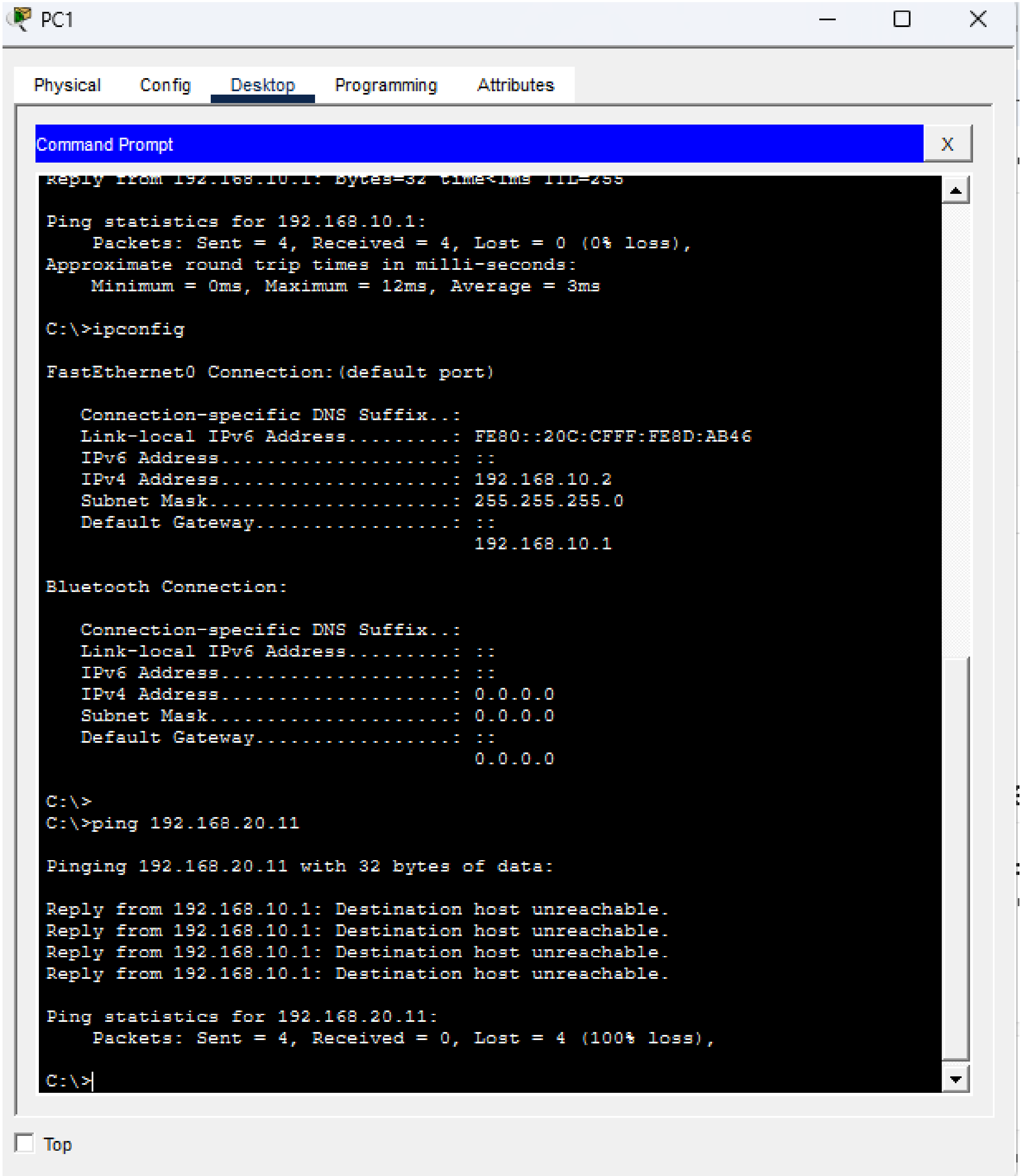

HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool VLAN10
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN20
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN30
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#
Copy
Paste
Top

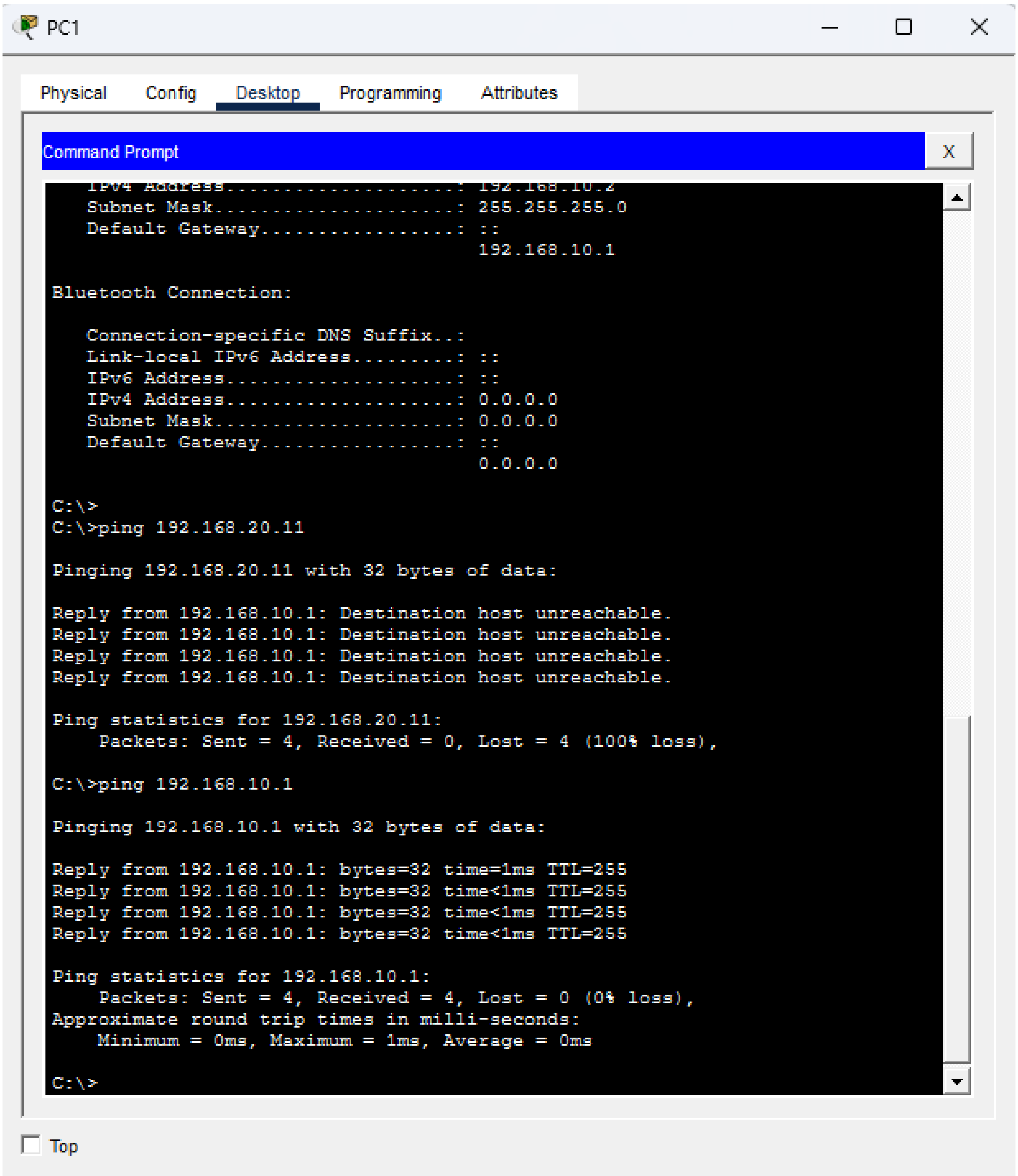


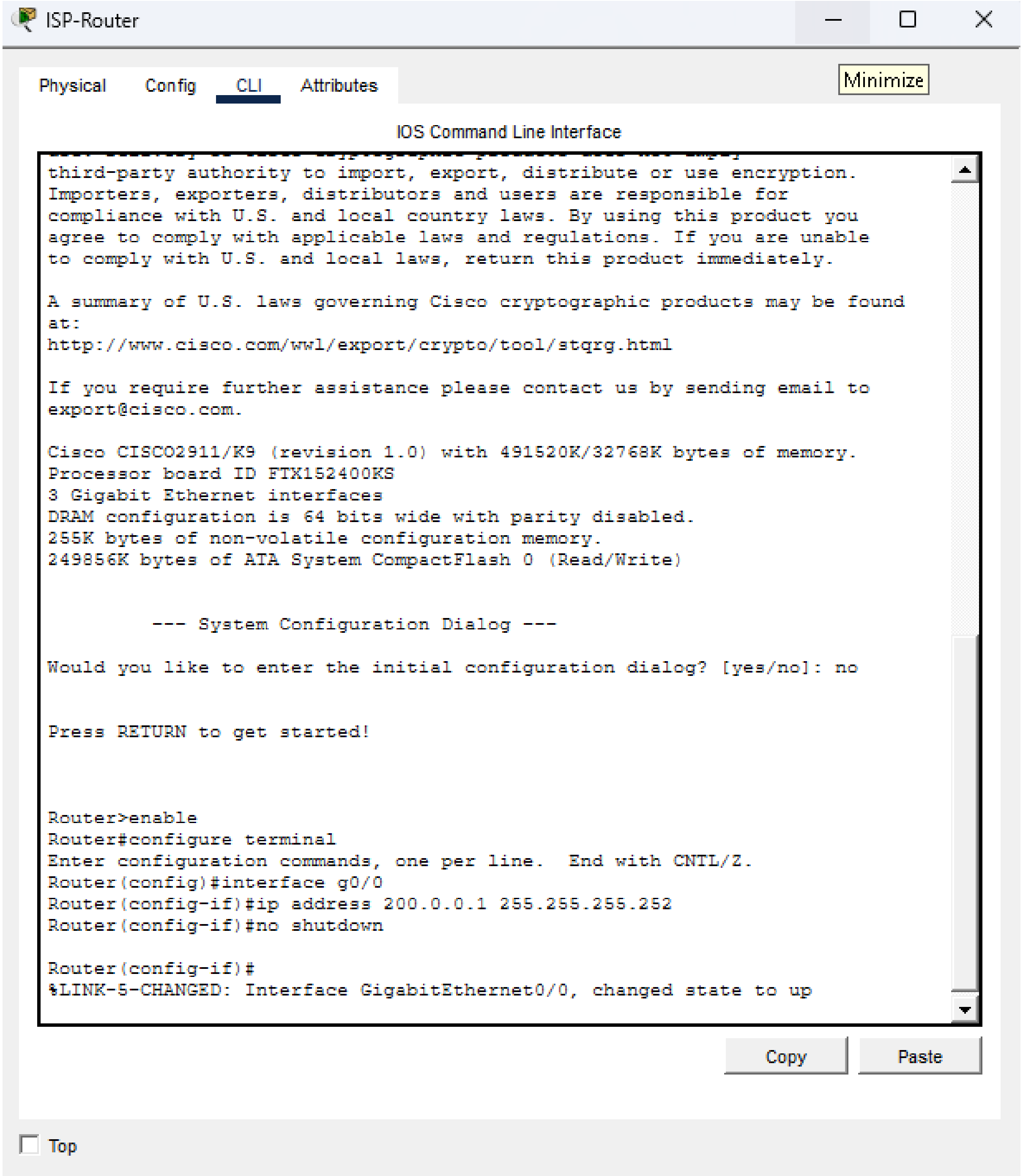

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#
01:28:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on Serial0/3/0 from
LOADING to FULL, Loading Done
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
O    192.168.10.0/24 [110/65] via 10.0.0.1, 00:00:35, Serial0/3/0
O    192.168.20.0/24 [110/65] via 10.0.0.1, 00:00:35, Serial0/3/0
O    192.168.30.0/24 [110/65] via 10.0.0.1, 00:00:35, Serial0/3/0

Router#
```


HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
% Invalid input detected at '^' marker.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip access-group 10 in
Router(config-if)#acess-list 100 deny ip 192.168.10.0 0.0.0.0.255
^
% Invalid input detected at '^' marker.
Router(config-if)#access-list 100 deny ip 192.168.10.0 0.0.0.0.255
^
% Invalid input detected at '^' marker.
Router(config-if)#exit
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.0.255
^
% Invalid input detected at '^' marker.
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255
% Incomplete command.
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255
% Incomplete command.
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0
0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface gigabitethernet 0/0.10
Router(config-subif)#ip access-group 100 in
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show access-lists
Standard IP access list 10
10 deny 192.168.10.0 0.0.0.255
20 permit any
Extended IP access list 100
10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
20 permit ip any any
Router#
Copy
Paste
Top

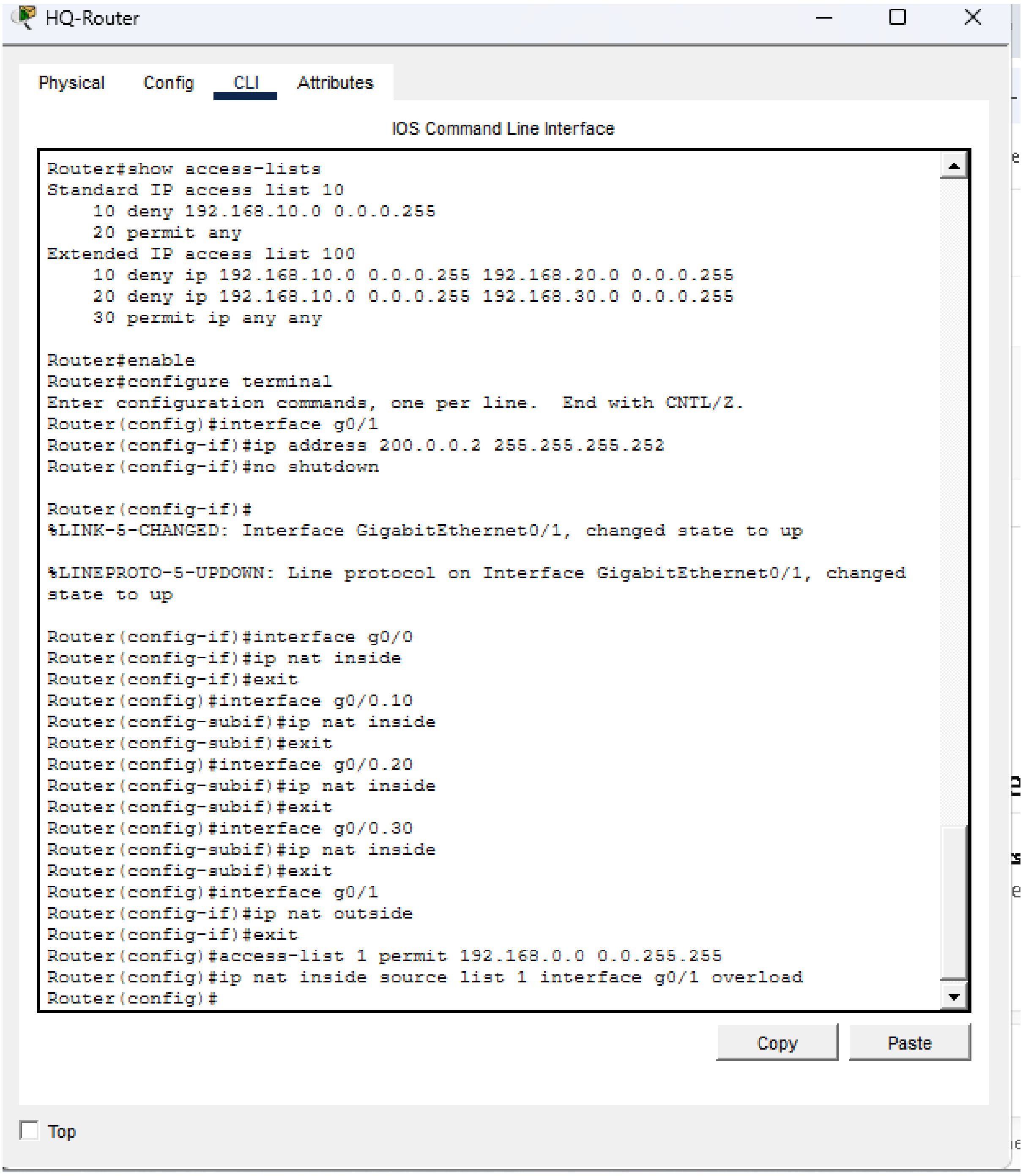

HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
10 deny 192.168.10.0 0.0.0.255
20 permit any
Extended IP access list 100
10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
20 permit ip any any
Router#show ip interface gigabitethernet 0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
Internet protocol processing disabled
Router#enable
Router#comfigure terminal
^
% Invalid input detected at '^' marker.
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no access-list 100
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0.10
Router(config-subif)#ip access-group 100 in
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show access-lists
Standard IP access list 10
10 deny 192.168.10.0 0.0.0.255
20 permit any
Extended IP access list 100
10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
20 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
30 permit ip any any
Router#
Copy
Paste
Top

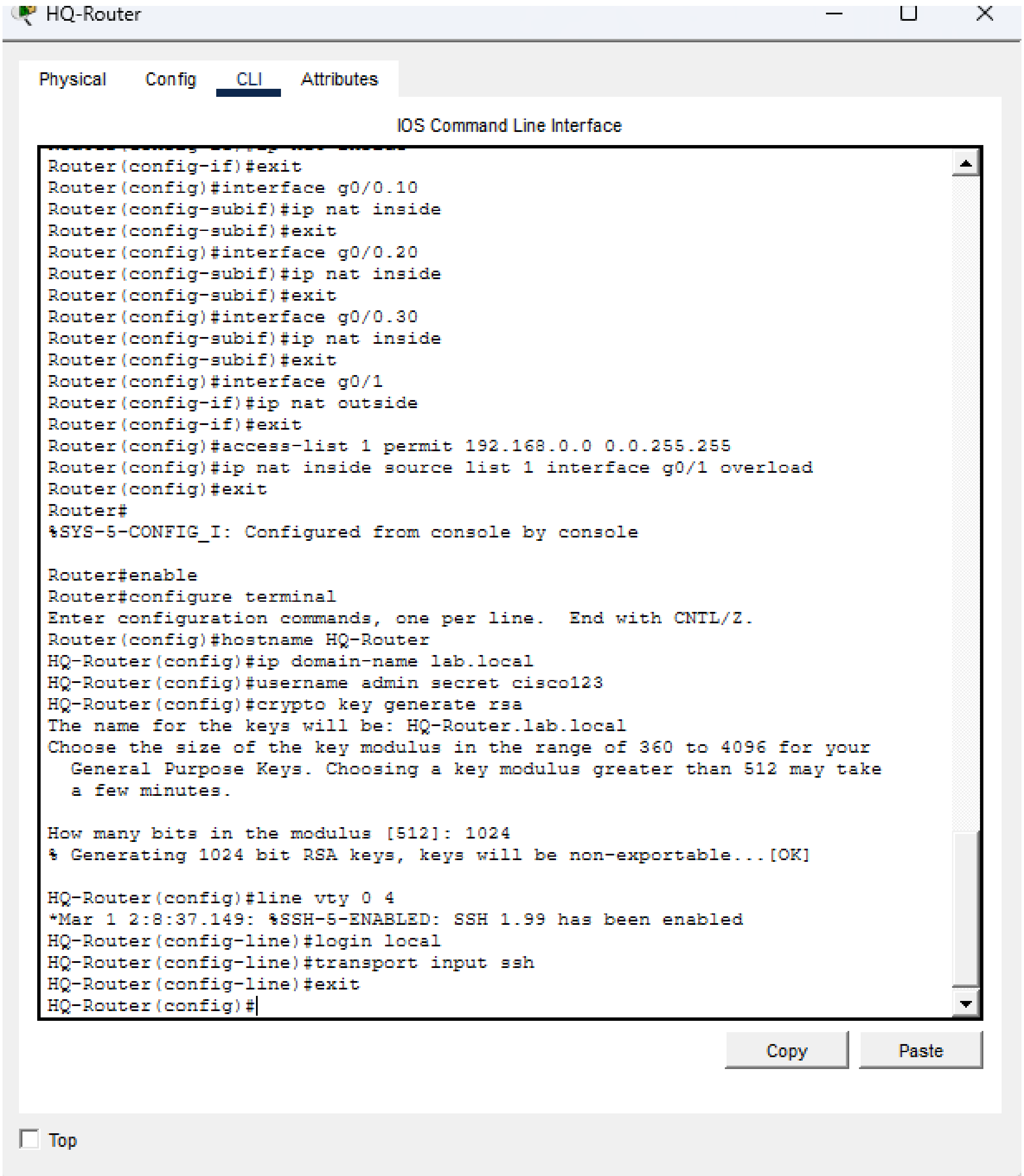

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 3ms
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: FE80::20C:CFFF:FE8D:AB46
IPv6 Address....................: ::
IPv4 Address....................: 192.168.10.2
Subnet Mask.....................: 255.255.255.0
Default Gateway.................: ::
192.168.10.1
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address....................: ::
IPv4 Address....................: 0.0.0.0
Subnet Mask.....................: 0.0.0.0
Default Gateway.................: ::
0.0.0.0
C:\>
C:\>ping 192.168.20.11
Pinging 192.168.20.11 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.20.11:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top

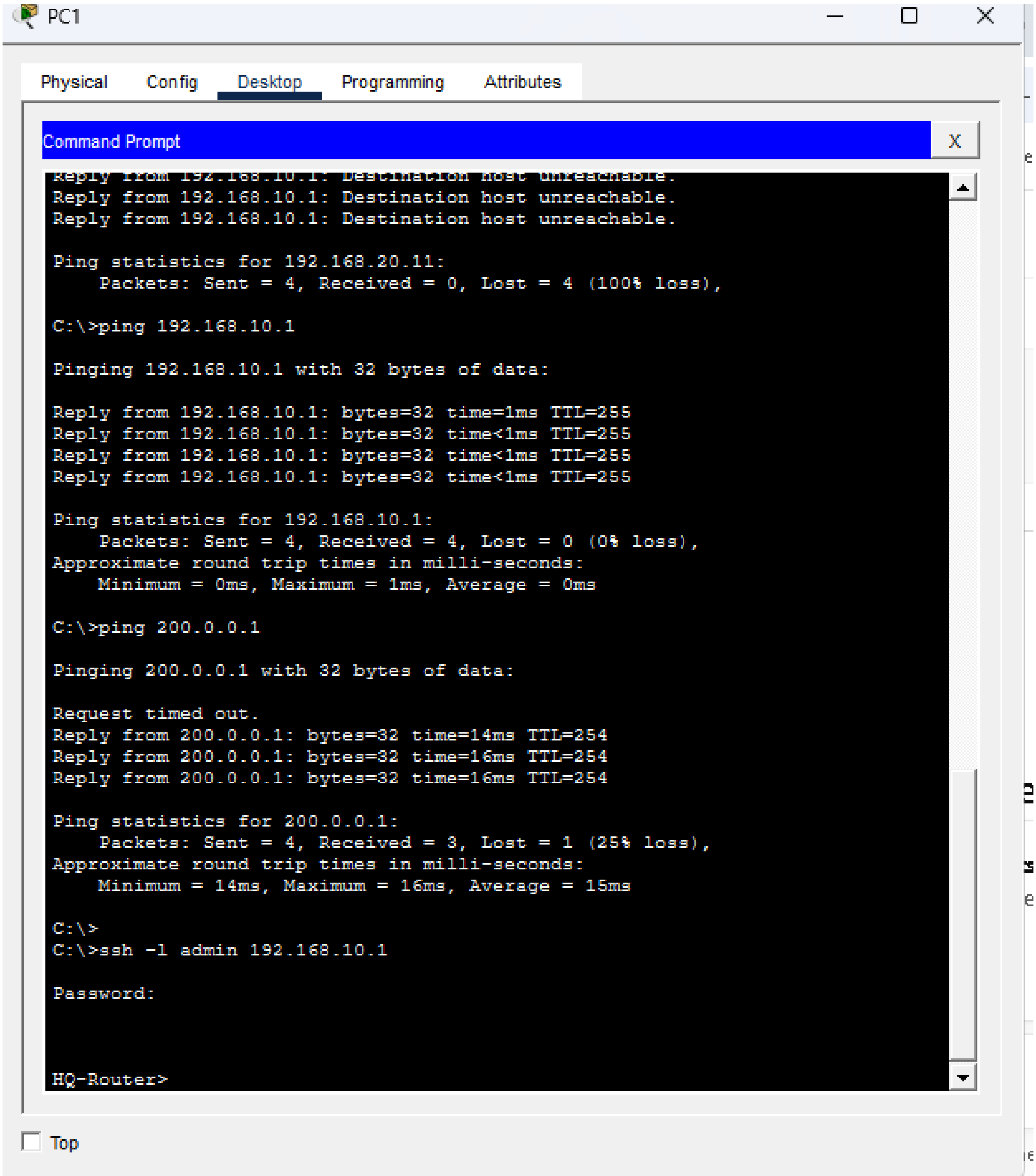

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
IPv4 Address......................: 192.168.10.2
Subnet Mask.......................: 255.255.255.0
Default Gateway...................: ::
192.168.10.1
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address......................: ::
IPv4 Address......................: 0.0.0.0
Subnet Mask.......................: 0.0.0.0
Default Gateway...................: ::
0.0.0.0
C:\>
C:\>ping 192.168.20.11
Pinging 192.168.20.11 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.20.11:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

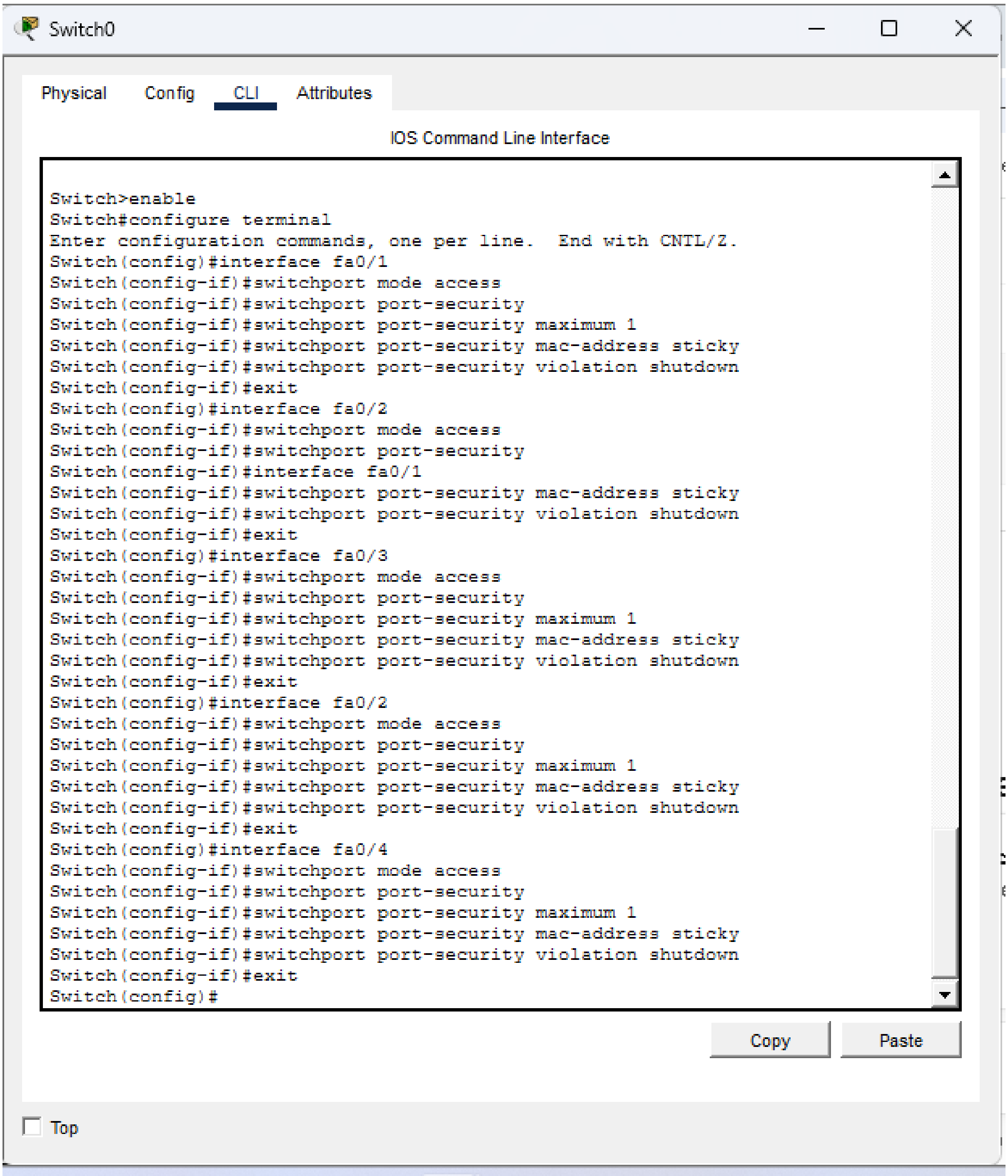



```
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found
at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 200.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```


HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#show access-lists
Standard IP access list 10
10 deny 192.168.10.0 0.0.0.255
20 permit any
Extended IP access list 100
10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
20 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
30 permit ip any any
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 200.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface g0/0.10
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface g0/0.30
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#
Copy
Paste
Top


HQ-Router
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#exit
Router(config)#interface g0/0.10
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface g0/0.30
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname HQ-Router
HQ-Router(config)#ip domain-name lab.local
HQ-Router(config)#username admin secret cisco123
HQ-Router(config)#crypto key generate rsa
The name for the keys will be: HQ-Router.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
HQ-Router(config)#line vty 0 4
*Mar 1 2:8:37.149: %SSH-5-ENABLED: SSH 1.99 has been enabled
HQ-Router(config-line)#login local
HQ-Router(config-line)#transport input ssh
HQ-Router(config-line)#exit
HQ-Router(config)#
Copy
Paste
Top


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.20.11:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 200.0.0.1
Pinging 200.0.0.1 with 32 bytes of data:
Request timed out.
Reply from 200.0.0.1: bytes=32 time=14ms TTL=254
Reply from 200.0.0.1: bytes=32 time=16ms TTL=254
Reply from 200.0.0.1: bytes=32 time=16ms TTL=254
Ping statistics for 200.0.0.1:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 14ms, Maximum = 16ms, Average = 15ms
C:\>
C:\>ssh -l admin 192.168.10.1
Password:
HQ-Router>
Top


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#interface fa0/1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#exit
Switch(config)#
Copy
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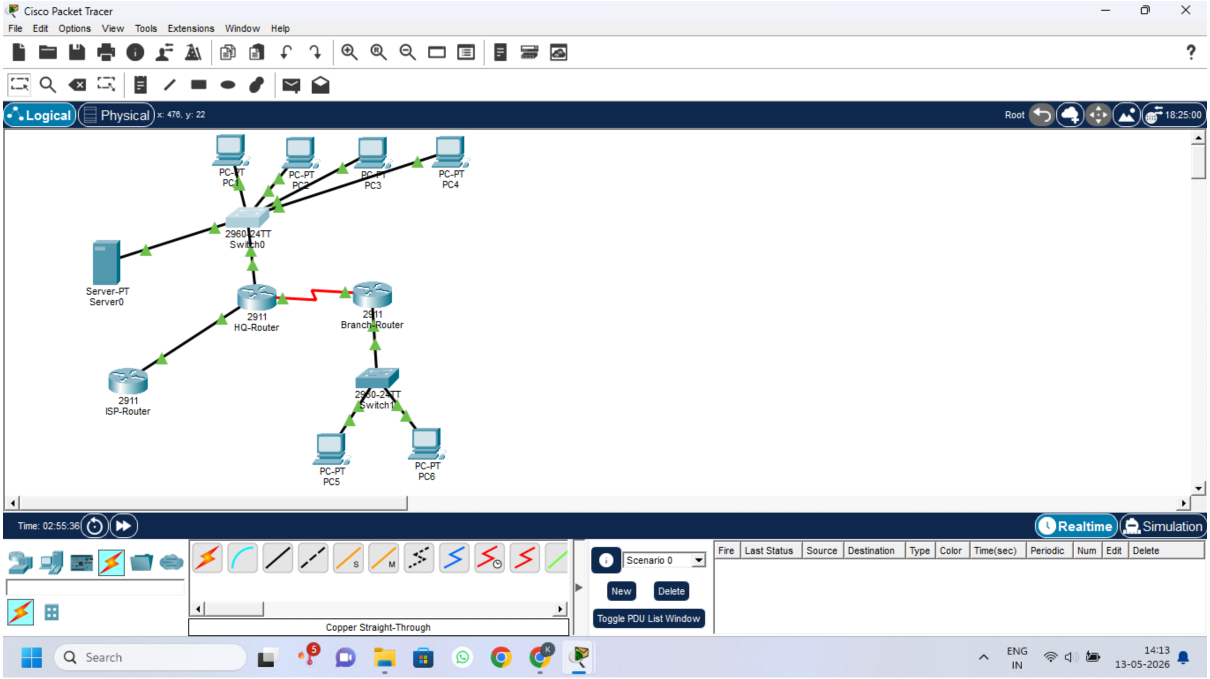
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 478, y: 22
Root
18:25:00
PC-PT
PC4
2960-24TT
Switch0
Server-PT
Server0
2911
HQ-Router
2911
Branch-Router
2911
ISP-Router
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6
Time: 02:55:36
Realtime
Simulation
Copper Straight-Through
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Search
ENG
IN
14:13
13-05-2026